

Trusting autonomous vehicles: An interdisciplinary approach

Kaspar Raats^{a,b,*}, Vaike Fors^b, Sarah Pink^c

^a Volvo Car Corporation, Gothenburg, Sweden

^b School of Information Technology, Halmstad University, Halmstad, Sweden

^c Emerging Technologies Research Lab, Faculties of Information Technology and Art, Design and Architecture, Monash University, Melbourne, Australia

ARTICLE INFO

Article history:

Received 11 June 2020

Received in revised form 13 August 2020

Accepted 15 August 2020

Available online xxxx

Keywords:

Human-computer interaction trust research

Intelligent automation

Autonomous vehicles

Emerging technologies

ABSTRACT

The theoretical concept of trust has been identified as highly important to the successful design of intelligent technologies such as autonomous vehicles (AVs). In human-centred transport research this has resulted in a focus on *trust* in the technical design of future AVs and has raised the question of how the conditions that form trust change as technologies become more intelligent. In this article we discuss the first stage of an interdisciplinary project that brought together ethnographic and experimental user studies into trust in intelligent cars. This stage focused on the development of an interdisciplinary methodological framework for the user studies, through a review of 258 empirical HCI research articles on trust in automation and AVs. The review investigated the following research questions: a) what are the key themes in HCI methodologies used to research trust in automation and AVs; b) how do they account for trust in AVs as part of wider contexts; and c) how can these methodologies be developed to include more than momentary and individual human-machine interactions. We found that while theoretical understandings of trust in automated technologies acknowledge the relevance of the wider context in which the interaction occurs, existing methodologies predominantly involve experimental studies in simulated environments with a focus on reliance related aspects of trust. We identified that ethnographic user studies can potentially contribute to new connections between theoretical understandings and conventional experimental methods. Therefore, we propose a framework for an interdisciplinary approach that combines experimental and ethnographic methodologies to investigate trust in AVs.

1. Introduction

Autonomous vehicles (AVs) are one of the most widely investigated and followed technologies within the automotive domain (Beiker and Calo, 2012). Vehicles are becoming increasingly automated through the introduction of intelligent features such as adaptive cruise control (ACC), lane keeping aid (LKA), and driver monitoring for inattention or health issues. However, fully autonomous vehicles are still not available as consumer products and there is growing public scepticism towards AVs (Hengstler et al., 2016). For instance, the American Automobile Association (AAA)¹ reports that only one-in-ten (10%) U.S. drivers would trust to ride in a fully self-driving vehicle and 28% are unsure. In this article, we lay out the foundations for an interdisciplinary approach that addresses this problem by combining ethnographic insights with model-based analysis to better understand how trust in AVs develops in the everyday context. Our argument is based on a systematic analysis of existing research that demonstrates how conventional methodologies within transport and Human-Computer Interaction (HCI) studies are not designed to target the complex embeddedness of trust in everyday life situations. Our review revealed the gaps in

knowledge and understanding that have come about because social and cultural understandings of why people develop trust in AVs are not sufficiently accounted for in existing approaches. It has been argued that this gap can be avoided through methodologies that add qualitative methods to clarify quantitatively gathered findings (Cohen et al., 2020; Creswell, 2009; Magnusson et al., 2020; Orlovskaya et al., 2020), and that use situated and contextual insights as indicators for what is of interest for further investigation (Chng et al., 2019; Diefenbach and Hassenzahl, 2019; Raats et al., 2019). In this article we develop this argument further by analysing how such an interdisciplinary approach would benefit research into trust in AVs and what knowledge gaps it would fill. The literature review research discussed here was developed to build an interdisciplinary framework for the Trust in Intelligent Cars (TIC) project between 2018 and 2020,² which was a collaboration between academia and the car industry.

User and public trust are seen as essential to the success of new intelligent technology design and therefore as also critical for the success of automated driving (Lee et al., 2015; Wintersberger and Riener, 2016). Thus, the concept of trust plays a crucial role in the theorisation of the acceptance of innovative technology (such as AVs) in HCI research (Lüders et al., 2017) and is understood to determine people's willingness to rely on automation in uncertain situations (Hoff and Bashir, 2015). Therefore,

* Corresponding author at: Volvo Car Corporation, Gothenburg, Sweden

E-mail address: kaspar.raats@hh.se (K. Raats).

¹ <https://newsroom.aaa.com/2020/03/self-driving-cars-stuck-in-neutral-on-the-road-to-acceptance/>

² TIC - Trust in Intelligent Cars

trust has emerged as a relevant focus in research on the implementation of innovative automated technologies in society (Hengstler et al., 2016). Car manufacturers aspire to understand how to engender both user and public trust and acceptance of AVs so that people will be more comfortable travelling in and sharing public roads with AVs. On the basis of their 2019 vehicle technology survey, the AAA suggest that key to helping consumers feel more comfortable with fully self-driving vehicles will be “bridging the gap between the perception of automated vehicle technology and the reality of how it actually works in today's cars”.³ These ambitions have oriented Human-Computer Interaction (HCI) researchers to investigate how people will adopt and accept AVs on public roads, and to develop theoretical models to understand this in terms of *trust* (Hoff and Bashir, 2015). In this article we present the results from the first work package within the TIC project. This work package aimed to establish the theoretical and methodological foundations for an interdisciplinary approach to trust in AVs which combined ethnographic research with experimental testing, and in doing so to determine a rationale and practical framework for empirical research. We undertook a systematic review of how HCI research methodologies have been developed within transport studies in response to the call for understanding user and public trust. The review investigated the following research questions: a) what are the key themes in HCI methodologies used to research trust in automation and AVs; b) how do they account for trust in AVs as part of wider contexts; and c) how can these methodologies be developed to include more than momentary and individual human-machine interactions. Before we address these questions through a presentation of our analysis of current research, we next discuss recent theoretical developments within HCI research that are relevant for framing an interdisciplinary approach.

1.1. Emerging HCI research agendas

New HCI research agendas have taken form in the wake of the development of artificial intelligent (AI) technologies and smart automation. Moving from traditional ideas that HCI is about engineering human-machine coupling or connectivity that needs to be optimised, later research related to increasingly mobile and tangible technologies opened up to understanding the social, cultural, emotional and embodied experiences of technology in a messy, real world outside of the rationalities of work and purpose. This line of thought has been called *third wave* HCI (Bødker, 2006; Dourish, 2001; Harrison et al., 2007). Recent literature has now identified a need for a *fourth wave* HCI to acknowledge how, with the advent of for example AI, social robotics, cyber-physical systems and automated vehicles, human-machine interaction has become increasingly fuzzy in terms of boundaries between humans and the technology (Frauenberger, 2019). The implementation of these technologies makes it harder to understand where the interface is that needs designing, since it is no longer self-evident where the human ends and the technology starts. Frauenberger (2019) argues that this shift demands new theoretical HCI research agendas based on “entanglement theories”, that “decentre the human as the sole source of activity and ... elevate the role of the non-human world from a passive backdrop to human activity, to active contributors to relational action as it unfolds” (Frauenberger, 2019, p. 21). Third and fourth wave HCI imply that people's interactions with intelligent technologies in AVs have to be understood in a wider context, that exceeds the momentary and individual human-machine interaction that underpins traditional HCI engineering culture. These approaches reflect what Hassenzähl (2010) argues has become the main focus within HCI research; to focus on “the ‘what’ and the ‘how’ of interaction”, thereby leaving the question of why people do things in the interactions out of scope.

In the automotive industry studies are undertaken early in the design process, prior to the release of a technology, to evaluate if a concept is potentially acceptable and usable, as well as after it has been launched onto the market. This works well in the case of specific features that can be implemented and evaluated in already existing cars. However, AVs that do not yet

exist are a different case. Through the shared agency between the car and the driver, AVs will change the way people experience driving and how they will interact with the car and the surroundings. Therefore, we need to investigate how the advanced machine intelligence that is required for fully autonomous vehicles will impact on how trust develops in the entanglement of the human and non-human actors in autonomous driving in real-world environments. Below we discuss how our research revealed the gaps in knowledge and understanding of trust in automation and AVs and highlight the importance of combining different research approaches to encompass not only the social, material, embodied and messy every-day world that the third wave HCI (Harrison et al., 2007) highlighted, but also to open up for investigations of what happens with trust when the “fourth-wave” technologies with their blurred human-machine interfaces start to inhabit it (Frauenberger, 2019). In the next section we present our analysis of the state of the art HCI transport research on understanding trust in automation and AVs. Following that we return to the emerging research agendas of fourth wave HCI, which provide theoretical foundations for creating an interdisciplinary approach that takes accounts for the situational and contextual aspects of trusting intelligent cars as part of what is experienced. Indeed, as Fridman et al. (2019) argue, without an in-depth analysis of the interaction between driver and AV in various driving contexts, an assessment is incomplete.

2. Understanding trust in automation and AVs

In this section we present the first part of our analysis of the theoretical underpinnings of transport research on trust in automation. There is increased interest in trust in automation and AVs within the HCI research community (see Fig. 1). This growing number of publications indicates the importance given to trust in this research field. The wide interest has led to different definitions, factors influencing, models for studying, and guidelines for designing trust into the interfaces and behaviour of automation in general. One of the most common definitions of trust in automation, appropriated from interpersonal trust concepts, is “the attitude that an agent will help achieve an individual's goals in a situation characterized by uncertainty and vulnerability” (Lee and See, 2004), cited or relied upon by for example Hoff and Bashir (2015), Wintersberger et al. (2017) and Edelmann et al. (2019).

Three models of trust stand out in the evolution of the study of trust. First, Muir (1994) illustrates the trust relationship between a human and a machine elaborating on the individual's mental model governing trust formation, emphasizing that bases of trust change through experiences with a system and over time. Second, Lee and See (2004) illustrate the dynamic process that rules trust and its influence on reliance, introducing context as an important factor in trust development. Third, Hoff and Bashir (2015) discriminate between trust formation before and during an interaction. They introduce three layers of trust identified by Marsh and Dibben (2005): dispositional, situational, and learned trust. They specify dispositional factors such as culture, gender, age and personal traits; suggest that trust depends on the context of interaction; and elaborate on the dynamic nature of trust by distinguishing between initially learned trust and the dynamic trust that evolves through experience. They illustrate how design features indirectly influence learned trust through system performance, and point out factors such as the level of effort required to engage with a system, time constraints, situational awareness and well-being of a user, and alternatives to using a system that can guide reliance without necessarily influencing trust (see Fig. 2).

The understanding of trust in automation and AVs within HCI research has developed towards becoming more and more interested in how factors outside of the specific human-machine interaction situation, beyond questions of solely usability and perceived use, have an impact on what makes people comfortable with a technology at hand (Hassenzähl, 2010). Hoff and Bashir (2015) argue for studying trust influencing factors in real-world environments and point out the need to investigate:

- how culturally different knowledge and expectations towards automation can influence trust and reliance behaviours;

³ <https://newsroom.aaa.com/2019/03/americans-fear-self-driving-cars-survey/>

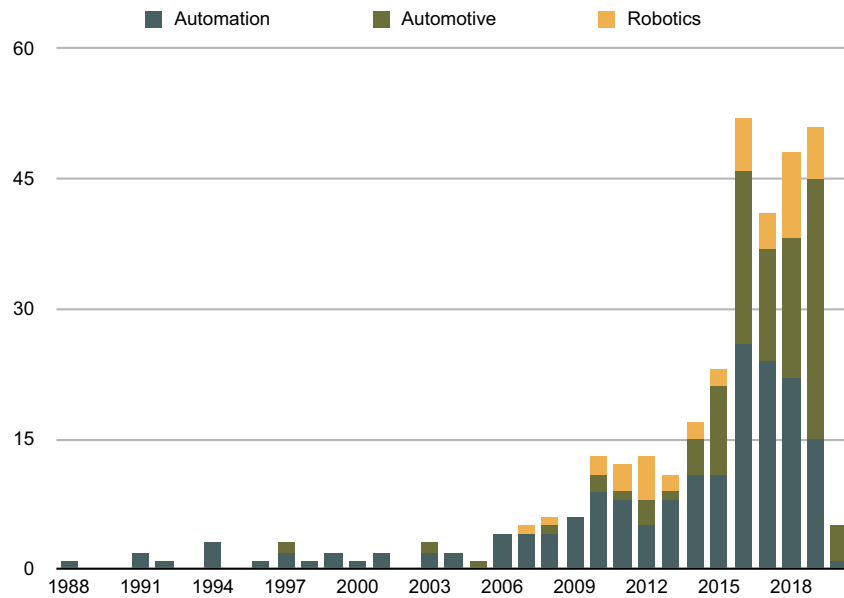


Fig. 1. Literature search results categorised into automation, automotive and robotics, and how they spread over the years.

- how trust in automation develops in people from different nationalities, religions and ethnicities;
- how context dependent internal factors (e.g. mood, self-confidence) guide trust formation;
- how stress, boredom, energy levels, and task motivation influence trust in automation; and
- how different relationships between previously listed characteristics would guide trust development.

In summary, the dominant understanding of trust has developed from mainly being concerned with what affects a single operator when interacting with a technology to more recently include a multidimensional construct guided by dispositional, situational, and learned factors, including how trust changes through the experience with a technology or a system. Indeed, as the recent models of trust indicate, trust is something that people feel and not only rationally reason about, it is influenced by personality traits and cultural context and it evolves through experience. We next

explore how the theoretical development described above has been mirrored in HCI research. We examine how it is articulated methodologically, which aspects of trust have been explored, and where arguments are made as well as gaps emerge that indicate a growing interest in and need for trust research that investigates factors beyond specific human-machine interaction situations. In the next section we outline our literature review and analysis of research focusing on trust as either its main line of inquiry or within a wider research agenda.

3. Researching trust in automation and AVs

In this section we present the results of our review and analysis of 258 empirical HCI research articles. We report on which methodologies are being used, which aspects of trust are investigated in HCI-research on automation and how these influence the ways in which research into trust in autonomous vehicles accounts for experiential, social and cultural factors. To incorporate the influence of research into trust in automation more

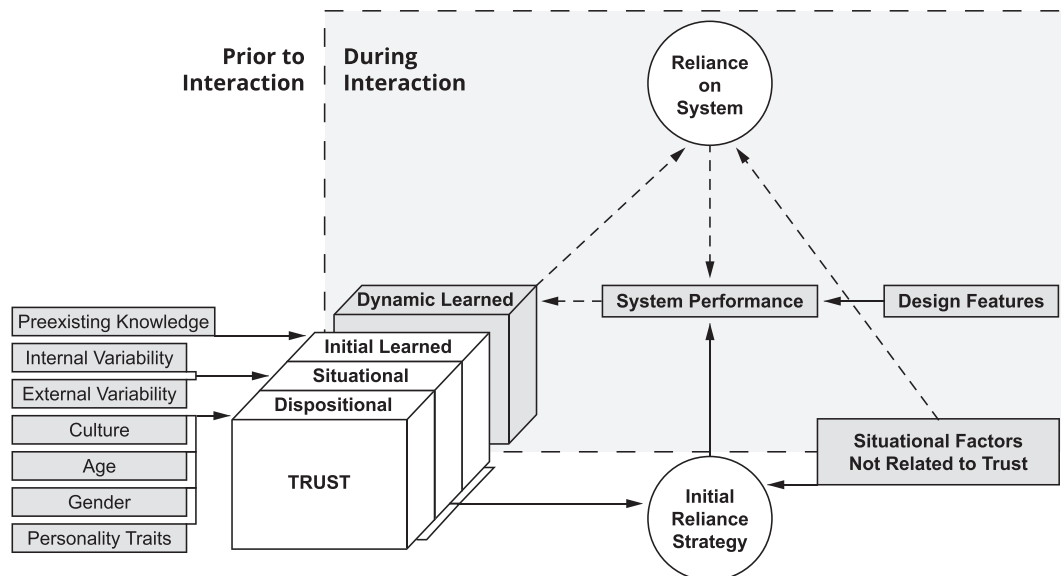


Fig. 2. Full model of factors influencing trust in automation by Hoff and Bashir (2015).

widely on research about trust in AVs as a form of intelligent automation (Hengstler et al., 2016) our review covered literature on both trust in AVs and trust in automation. The review process was guided by the PRISMA Statement (Moher et al., 2009) that defines a procedure for meta-analysis.

3.1. Process

3.1.1. Source and search string

The review discussed in this article was undertaken in November 2019, using five databases: Engineering Village, IEEE Xplore, ScienceDirect, Scopus and Web of Science. A query string consisting of main terms generated from the keywords used in existing HCI trust in automation and AVs literature was created. The string “trust* AND (‘human automation interaction’ OR ‘human centred design’ OR ‘user centred design’ OR ‘human computer interaction’ OR ‘user experience’ OR ‘human factors’ OR ‘human engineering’) AND (‘autonomous vehicle’ OR ‘automated vehicle’ OR ‘unmanned vehicle’ OR automation OR driving)” was then used to search from the title, abstract and keywords field. Exceptions were IEEE Xplore where we searched from all metadata and Engineering Village where the fields were title, abstract and subject. The query resulted in 1709 total records (565 from Engineering Village, 336 IEEE Xplore, 31 ScienceDirect, 522 Scopus, 255 Web of Science).

3.1.2. Selecting the studies

To identify the key themes used to research trust in automation and AVs in HCI methodologies we selected studies focusing on trust as either their main research question or studies that investigated trust within a bigger research agenda. We disregarded studies which reported only on the implications for trust as an outcome of research with a different focus as these studies did not contribute to discussion of the theory or methodology for investigating trust in transportation research.

First, we exported the meta-data of the results as comma-separated values (CSV) or text files (TXT). In many cases this created multiple files as the database allowed a limited number of records to be exported at one time. In case of multiple export files we created a separate MS Excel file for each database to merge the records. We then removed any empty values and saved the cleaned data as CSV files. This process removed eight entries, leaving a total of 1701 records. These data were then merged into a single table by importing them into the Ninodb database. Due to differences in the field naming conventions of the different databases (e.g. Publication year, Year) we imported the data from the Scopus database first and merged the data from other databases using the Scopus categories.

Second, in Ninodb we removed duplicate entries and filtered the results to include only journal articles, early research articles, conference papers, book chapters and books. This led to 1041 distinct records.

Third, on the basis of the abstracts we identified 1) papers focussing on trust as the subject of inquiry (rather than having findings on trust as a

result of their analysis) and 2) articles where the abstract didn't clarify the focus. This narrowed the results down to 375.

Fourth, we read the articles and rejected 46 where trust was not a central research focus. This left a final number of 329 records (see Fig. 1).

3.1.3. Categorisation

Next, we categorised these 329 records as either: empirical studies investigating trust or aspects of trust through experiments; or theoretical articles proposing a trust framework, a model, a questionnaire, or conducting a literature review of trust research. Within these categories we distinguished between trust research focusing on automation (e.g. automated control systems, automated decision aids), automotive (e.g. self-driving cars) or specifically in the field of robotics (e.g. human-robot interaction, robot agents). We identified 71 theoretical papers (45 in automation, 15 in automotive, and 11 robotics) and 258 empirical studies (132 in automation, 93 in automotive, and 33 in robotics).

3.2. Analysis: methodologies and aspects of trust

The analysis of 258 empirical articles showed that the applied research methodologies vary, but dominantly rely on research conducted in a laboratory setting using questionnaire based trust assessment. Table 1 provides a summary of different research contexts and methodologies. We used Hoff and Bashir's (2015) trust model (see Fig. 2) to analyse which aspects of trust the different reported studies investigated.

In the analysis of the literature we paid particular attention to the inter-relationship between how the studies were designed in terms of the context in which the research was conducted, the research methodologies used in these contexts and how these approaches acknowledge the dynamic and the situated nature of trust advocated by Hoff and Bashir (2015) and Lee and See (2004).

3.2.1. Study context

Of the 258 empirical investigations (see Table 1), most, 209 (81.0%), were conducted through simulated scenarios in laboratory setups, test tracks or using computer gaming scenarios. Fig. 3 illustrates the development of this trend in conducting laboratory setups as the main or accompanying research approach for studying trust in automation in AVs, in relation to other contexts. Study participants involved in research in these experimental settings do not necessarily subsequently use the systems tested in real-life situations. These are followed by online studies, of which there were 29 (11.2%) in which participants were approached over the internet and asked to fill in a survey. 12 (4.7%) of the investigations were held in real-life environments, six (2.3%) involved field studies with people in their everyday life situations and five (1.9%) were on-road studies on public roads in real-traffic. Of the total empirical investigations one (0.4%) was a multiple case study analysis, which explored empirically how companies foster trust in autonomous vehicles and medical devices.

Table 1

Summary of research context, data collection methods and how they connect to Hoff and Bashir's (2015) trust model.

Method		Trust model						
		Initial learned	Dynamic learned	Dispositional trust	Situational trust	System performance	Design features	Reliance on system
Study context	Field study		1	2	2	2	2	1
	Multiple case study analysis	1						
	On-road study		2			1	3	1
	Online study	7	3	12	4	12	10	3
Data collection	Simulator / Test track / Laboratory / Game	31	24	21	39	77	90	25
	Questionnaire / User feedback	40	24	30	42	77	89	27
	Physiological measurement		2		2	1	3	
	Observation	1	1	3	1	3	4	3
	Interview	6	4	3	2	4	14	1
	Focus group	1					1	
	Eye-tracking	2	2	1	1	2		
	Ethnography			1				
	Analysis of videos						1	1

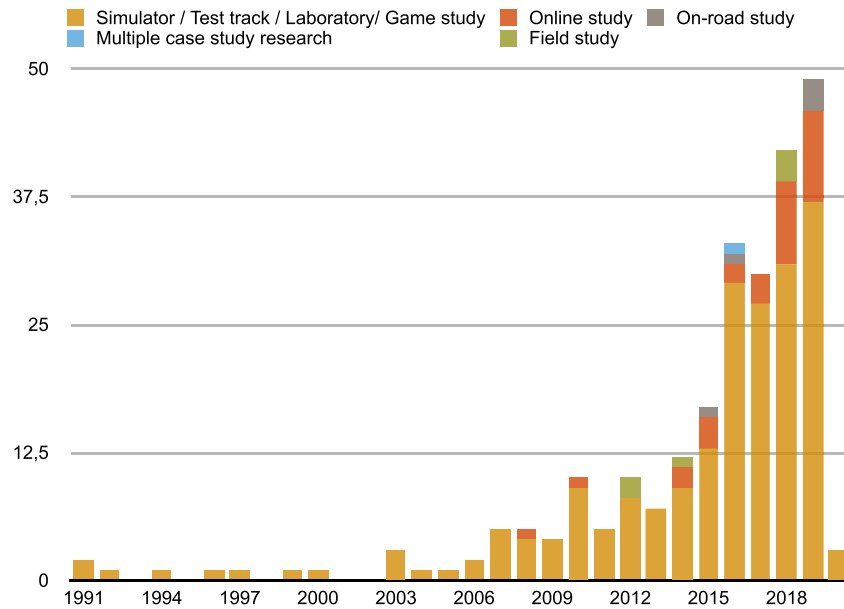


Fig. 3. Study context's chosen for trust research and how they distribute yearly.

3.2.2. Data collection

The majority of the 258 empirical investigations (see Table 1), 221 (84.9%) used questionnaires as a method of inquiry. These studies tended to use established questionnaires, predominantly employing the Trust in Automated Systems scale (Jian et al., 2000). Fig. 4 illustrates the development of this trend in questionnaire use as the main or accompanying data collection method for studying trust in automation in AVs, in relation to other methods. Questionnaires were followed by mainly semi-structured interviews in 30 studies (11.6%). 12 (4.7%) of the studies used observation as a method of data collection. Observations were conducted in real-life environments and in laboratory settings. Eye-tracking was used in six (2.3%) studies to track participants gaze behaviour as an indicator of trust. Physiological measurements were undertaken in five (1.9%) cases, measuring

trust through skin conductance response, heart rate change or brainwave activity. In four (1.6%) studies researchers analysed pre-recorded videos to detect levels of trust through participants' posture or glance behaviour during driving while performing a secondary task. Focus group sessions were conducted in two (0.8%) cases, where participants were engaged in an open discussion about their experience with an interface as well as commenting on each other's observations. In both cases the participants were experienced drivers either with specialised vehicles or with an intelligent feature in their personal car. From the total empirical investigations one (0.4%) allowed unforeseen insights to be generated by undertaking an ethnographic approach to exploring a new phenomenon in participants' own environment and situations. These methods are not necessarily mutually exclusive.

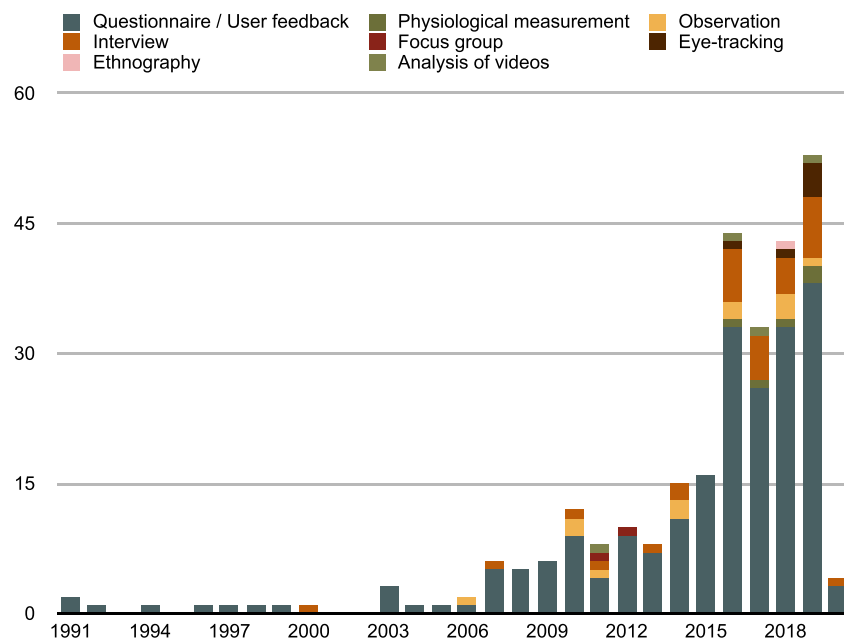


Fig. 4. Data collection methods used for trust assessment and how they distribute yearly.

3.2.3. Aspects of trust

Our analysis demonstrates that 256 (99.2%) of the total studies explored aspects that influence trust during interaction with a system. Fig. 5 illustrates that the most dominantly investigated aspects of trust in HCI research are design features, represented in 104 studies (40.3%), for example anthropomorphic features, transparent interfaces or system feedback properties, and system performance such as reliability, predictability or dependability in 91 studies (35.3%). These are followed by dynamic learned trust in 30 studies (11.6%) between a single operator and a system, and 29 (11.2%) studies that evaluate the reliance on a system during interaction. Aspects that influence trust prior to interaction with a system were investigated in 126 (48.8%) studies. More specifically, initially learned trust factors such as attitudes and expectations, or how prior experience with a system influences trust were investigated and situational trust factors such as self-confidence, mood, or perceived risks with relying on a system were investigated in 45 (17.4%) studies, seven times being investigated within the same study; and 36 (14.0%) studies explored dispositional factors, such as how cultural characteristics or age and gender influence trust. The different aspects of trust are often investigated together.

4. Taking trust research beyond human-machine interaction

Our purpose with this literature review was three-fold. As outlined above it sought to identify a) the key themes in HCI methodologies used to research trust in automation and AVs; b) how these methodologies account for trust in AVs as part of wider contexts; and c) how they methodologies can be developed to include more than momentary and individual human-machine interactions, and thereby open up routes for an interdisciplinary integration of ethnographic and experimental approaches. Below we first report on the findings of the review regarding how research on trust has been conducted to date, followed by a discussion of methodologies used in studies that have aimed to investigate trust in real life conditions. We end with a concluding section that demonstrates the implications of this for building an interdisciplinary approach for researching trust in automated vehicles.

4.1. Key methodologies

Our results highlight that the research methodologies used for studying trust in automation and AVs, tend to rely on quantitative laboratory research

(209 of 258, 81.0%) and questionnaire (219 of 258, 84.9%) based trust assessment. This becomes even more apparent when comparing the number of controlled laboratory research studies with studies conducted in real-world environments, to investigate different trust influencing factors in response to Hoff and Bashir (2015) trust model (see Fig. 6). Similarly, when comparing data collection methodologies, controlled questionnaire based trust assessment predominates over methodologies such as interviews, observations, focus groups and ethnography, that are designed to capture the social, situational and cultural aspects of trust from real-world environments (see Fig. 7). Figs. 6 and 7 illustrate that amongst the total empirical HCI research on trust in automation and AVs reviewed, controlled laboratory research and questionnaire based trust assessment were clearly the preferred methodologies. These approaches reflect what Hassenzahl (2010) argues has become the main focus within HCI research; to focus on “the ‘what’ and the ‘how’ of interaction” and thereby leaving the question of why people do things in the interactions out of scope, and limiting the extent to which the context in which interactions occur in is accounted for.

4.2. Method development

The theoretical understandings of trust in automation and AVs within HCI research have moved beyond questions of usability and perceived use to turn our attention towards how factors external to a momentary human-machine interaction impact on what makes people comfortable with a technology (Hassenzahl, 2010). In relation to research on trust and automation, Hoff and Bashir (2015) emphasize the need to investigate trust influencing factors in real-world environments and also explore dispositional and situational factors that influence trust. This means moving beyond the individual human-machine interaction to also include social, material, emotional and cultural factors that define the situation in which the interaction occurs. Hoff and Bashir's (2015) trust model (see Fig. 2) reflects how different relationships between these factors would guide trust development. This development calls for new methodologies designed to capture these complexities, and that examine not simply the moment in which trust occurs but also how and why it develops in specific forms in particular situations.

In the articles reviewed we analysed which of Hoff and Bashir's (2015) aspects of trust were investigated, and how the studies were designed in terms of data collection and research context. As shown in Table 1, and discussed above, the majority of the studies were performed according to

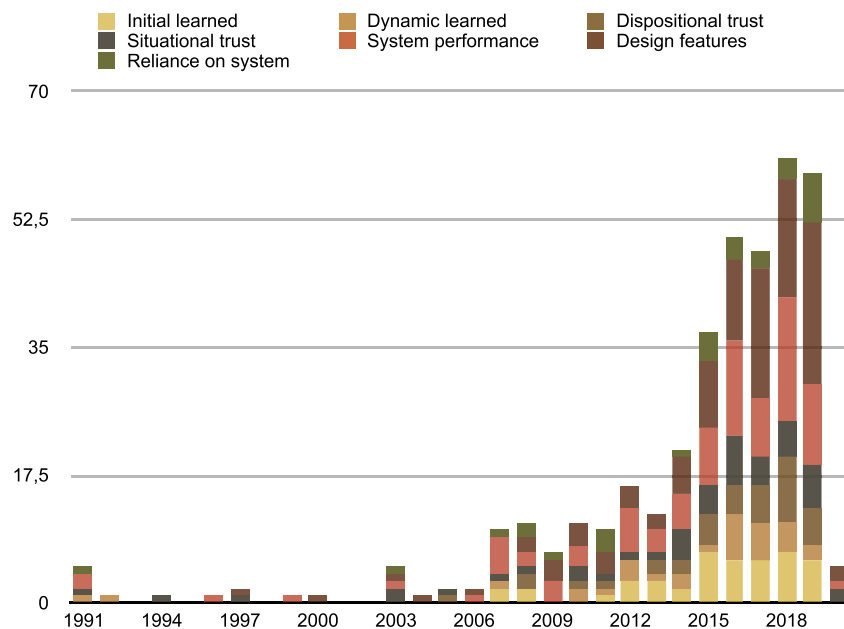


Fig. 5. Aspects of trust being studied and how they distribute yearly.

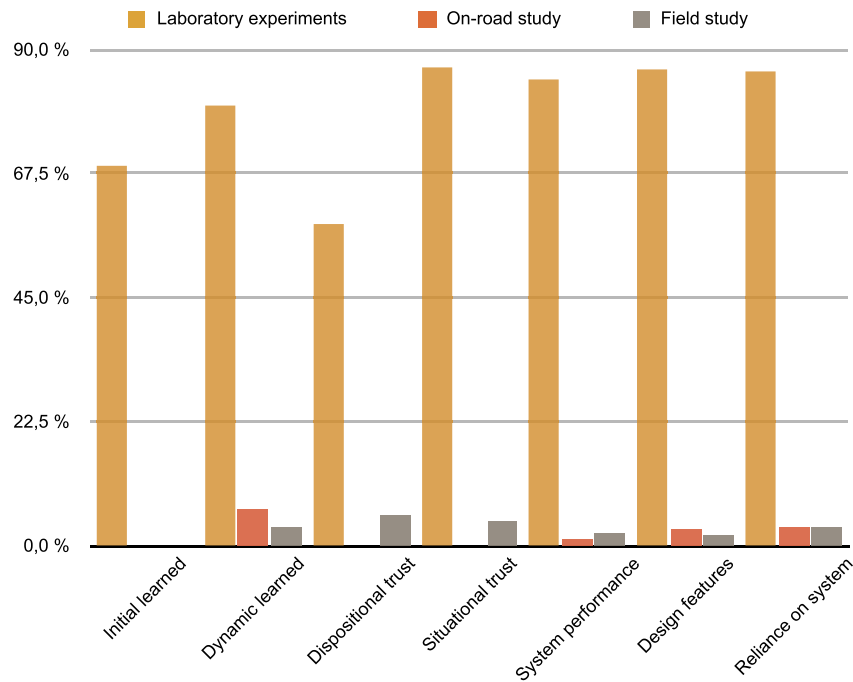


Fig. 6. Comparison of the aspects of trust addressed by controlled laboratory research studies and studies conducted in real-world environments.

controlled protocols and controlled environments to evaluate performance, reliance and design features. To better understand how methodologies have developed in response to the call for more complex understandings of the dispositional and situational aspects of trust, we analysed the studies that are conducted in real-life contexts. These studies used questionnaires, interviews, observations, focus groups and other ethnographic methods. Our analysis revealed two categories, differentiated in relation to the controls under which data collection was conducted.

The studies included in the first category were performed in real traffic, and used controlled trust rating questionnaires and semi-structured interviews as well as measurement methods. For example, studies in this category were designed to measure, range anxiety (Jung et al., 2015), the

regaining of control over AVs (Banks and Stanton, 2016), HMI designed for persons with impaired vision (Brinkley et al., 2019), and skin conductance while driving in AV's (Mühl et al., 2019). The main arguments for conducting the studies in real traffic instead of simulated situations were to add to the immersiveness of the drive and to increase the ecological validity. Robotics research has also moved into real-life contexts, where for example Piasek and Wieczorowska-Tobis (2018) investigated human-robot interaction between elderly people and social assistant robots in their homes, by asking them to fill in a questionnaire once a week over a 10 week period.

In the second category, methods have been exploratory, in the form of enabling participants elaborate on their real-life experiences. For example,

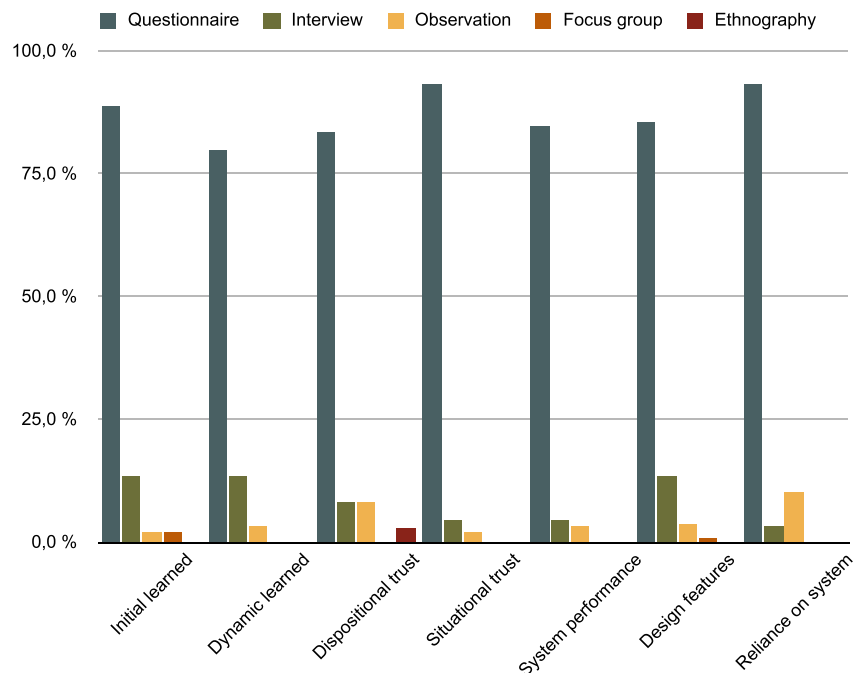


Fig. 7. Comparison of data collection methodologies and the aspects of trust they address.

Trösterer et al. (2014) studied owners of newly bought cars for eight weeks to learn how driver acceptance and User Experience (UX) evolved over time to understand trust in a Parking Assist System (PAS). They did not accompany participants in their real everyday environment but triggered events where the participants reconstructed their previous day and revived their real in-situ experiences. Observation techniques have been developed by Banks et al. (2018) in their study of Tesla drivers on real roads, involving unstructured observations to investigate the situations that occurred under real-life conditions while in automated driving mode. In the same vein, Balfe et al. (2018) observed experienced rail operators in four signaling centers, and analysed their activities in relation to their reported trust levels. McLoughlin et al. (2018) conducted open ended in-depth interviews in participants' homes to investigate how the needs and behaviours of elderly drivers change in their senior years, and how AV systems and services could be designed for these requirements. These methods have also been developed through longer-term ethnographic approaches, for example, Gridling et al. (2012) ran a two month ethnographic study with car-sharing platform users to learn about social and collaborative in-car activities between the drivers and passengers while driving. Through commuting together with participants and undertaking observations they were able to reveal how the behaviour of the drivers and passengers varied between different contextual situations as well as in relation to gender. Stahl et al. (2012) spent four weeks in gold mines to understand operators' perspectives on an automated system designed to optimize operational efficiency and productivity of haul traffic. Motivated by an earlier study that identified problematic interaction between haul truck operators and dispatch interfaces, focus groups and questionnaires were used to understand what caused a general attitude of suspicion towards the system. Instead of controlled questionnaires, these studies used open-ended observations and in-depth interview methodologies to capture the richness of the phenomena. Thus allowing participants the freedom to impart experience and evidence in their own words and language, and account for circumstances and surrounding context.

In sum, there is an emerging body of research that has identified a need to move beyond the focus on momentary human-machine interaction in simulated settings to uncover previously unknown aspects of why people develop trust in automated technologies. Balfe et al. (2018) argue that since almost all previous "experiments were laboratory based, utilizing simulations of work environments" the question if "these results transfer to real work environments is still an open question - but an important one given the increasing levels of automation in the workplace" (Balfe et al., 2018, p. 1). In the next section, we will summarize what can be learned from methodologies that move beyond controlled experiments, and how this can contribute to building interdisciplinary approaches.

4.3. Learnings from real world contexts

In our literature review we analysed trust in automation and AVs to answer the question of how trust in AVs has been researched within the field of HCI thus far, with the ambition to identify what methodological developments are needed to most effectively respond to recent extensions of theoretical trust frameworks. In the previous section we presented methodologies that were developed to respond to understandings of trust as evolving through the complexities of real life situations. In this section we pinpoint how studies represent the benefits of the real life approach compared to controlled experiments, and what we can learn about trust in automation from this approach that cannot be discovered through conventional experimental approaches. Our review suggests two major benefits of a real life approach can contribute to, and as Balfe et al. (2018) suggest, validate, results from experimental studies in simulated environments: studies undertaken in real life situations can reveal the unspoken details of what affects trust in everyday life, and how these circumstances are felt and experienced.

The value of real life studies in revealing the unspoken details of what affects trust in everyday life is demonstrated in two key studies. Stahl et al. (2012) demonstrate how while operators working in a gold mine

said they were comfortable with automated systems at first sight, open ended in-context interviews revealed that the operators considered the automated systems' inability to account for contextual information as a form of system failure. Similarly, through open observations of rail operators (Balfe et al., 2018), researchers were able to conclude that even though the individual workstations that were used by the rail operators were similar there was considerable variation in signalers' behaviour. This led them to understand that variation was driven by the individuals rather than by the workstations. Their conclusion would have been impossible to come to without considering how these tasks were undertaken in a real life context.

The significance of how everyday life circumstances are felt and experienced is also demonstrated through real life studies. For example, the power of studying everyday life situations to better understand the implications for trust development is advanced in Gridling et al. (2012) study, which used an ethnographic approach to research the social dynamics of trust in car-sharing situations, which would have been hard to capture in simulated environments. By immersing themselves in commuting with participants and through observations they revealed how the behaviour of the drivers and passengers changed, depending on contextual situations. McLoughlin et al. (2018), explored older adults' everyday life contexts by allowing participants the freedom to impart experience and evidence in their own words and language within their everyday circumstances and surrounding context. They argue that this approach was needed to be able to capture the rich context of trust and understand how AV systems and services could be designed for this particular target group. In the next section, we outline the benefits of combining real-life based and experimental research approaches.

4.4. Researching trust through combined methodologies

As noted above, our literature review and analysis was developed in order to create the foundation for a project that entailed an interdisciplinary approach to researching trust in intelligent cars. The project aimed to combine experimental and ethnographic methodologies in AV studies on public roads by engaging both the strengths of laboratory experiments in hypothesis testing through rigorous control measures to pinpoint sole explanations to a phenomena (Tanner, 2018) and those of ethnographic approaches in capturing the rich explanation and meanings of people's everyday life interactions with technology (Balfe et al., 2018). In this section we integrate our findings from the analysis of the literature above to discuss how ethnographic and experimental research methodologies complement each other.

The analysis of the literature demonstrated that HCI research into trust in automation and AVs has built a significant body of knowledge about trust based on experiments conducted in simulated environments, with a key focus on how system performance and design features influence trust. Simultaneously, a growing body of research proposes moving beyond laboratory experiments and exploring the situational and dispositional aspects of trust development. According to Hassenzahl (2010) and Harrison et al. (2007), HCI trust research conventionally follows a logic rooted in traditional HCI engineering research agendas, that assume trust evolves between a person and a machine in a momentary interaction. In contrast, emerging research argues that trust is continuously ongoing and evolves through people's experiences with technology in contingent everyday life situations in socially and culturally specific circumstances (Pink et al., 2020). As we outlined in Sections 4.2 and 4.3 experimental and ethnographic insights complement each other within the framework of existing trust models, demonstrating how ethnographic (and other real-life based approaches) investigate aspects of trust that are not revealed by dominant experimental approaches. An approach that combines experimental and ethnographic research thus contributes to knowledge about how the dynamics of trust formation are embedded in situational factors, as part of people's real-life complex social encounters, contingent circumstances and improvisations with technologies. Thus, an ethnographic approach helps us to understand how, for example, the factors identified by Hoff and

Bashir (2015) as the “Situational Factors Not Related to Trust” (see Figs. 2 and 8) also participate in creating the context through which trust evolves. In Fig. 8 we have illustrated the former by adding a clarifying layer on top of Fig. 2 that shows which factors the different approaches can address and how combining the different approaches can contribute a more complete understanding of trust in AVs.

The implications of the analysis and conclusions above for the TIC project was to design a methodology that rather than chronologically ordering one approach before the other, integrated the different approaches and enabled them to feed into each other. Experimental studies involved participants driving WOz cars to assess and measure trust through fixed questionnaires, video and audio analysis, and biometric data collection. Exploratory investigations involved ethnographic fieldwork and in-depth interviews to investigate how intelligent technology influences trust development in participants' everyday lives. Three iterations of this process allowed the experimental methods to evolve by taking into account insights from the ethnographic fieldwork while insights from the experiments were used to design ethnographic studies to validate the measurement findings in real-life situations (Balfe et al., 2018).

We acknowledge that our review was limited to HCI-research on trust in automation and AVs, and therefore does not account for the theories of trust from which the approaches discussed earlier originated. Furthermore, we are aware that the concept of trust in HCI is connected also to other concepts including acceptance and adoption. Due to the focus and scope of this article we do not elaborate on these relationships.

5. Conclusion

The purpose of this article was to: identify key themes in contemporary trust research methodologies in HCI; understand the extent to which these methodologies account for the contextual elements of how trust in AVs occurs; and establish how these methodologies can be developed to expand beyond a focus on momentary and individual human-machine interaction through an interdisciplinary approach that combines experimental and ethnographic insights. On the basis of this, the article lays out the foundations for an interdisciplinary approach.

Our analysis of 258 empirical HCI research articles on trust in automation and AVs found that research methodologies used for studying trust in automation and AVs, tend to rely on quantitative laboratory research and questionnaire-based trust assessment. HCI trust research is conventionally undertaken according to a logic that trust evolves between a person and a machine in an interaction situation that resonates with traditional HCI engineering research agendas (Harrison et al., 2007; Hassenzahl, 2010).

We also found that other methods have been brought into the trust research field in response to recent calls for approaches designed to investigate the complex embeddedness of trust in everyday life by accounting for social and cultural understandings of why people think and act with technology (Cohen et al., 2020; Dourish, 2001) or as Hassenzahl (2010) puts it moving beyond only answering, “the ‘what’ and the ‘how’ of interactions” to focus also on the ‘why’ people do things in the interactions. However, few, if any, of the reviewed methodologies were designed to capture the new social and technological complexities that artificial intelligence (AI) and smart automation enable in the form of what have been referred to as entangled interfaces between human and non-human actors (Frauenberger, 2019).

Therefore, to develop this body of knowledge we propose building on the interactional theories of trust that inform existing quantitative and testing research, by engaging with social theories of trust that account for the wider social and cultural context in which the interaction occurs (Corsín Jiménez, 2011). We present an interdisciplinary research methodology that not only combines experimental and ethnographic research methodologies, but also integrates them through ongoing exchanges of insights through the life of the project. In doing so, we call for the development of an interdisciplinary trust and automation research agenda that encompasses not only the social, material, embodied and messy every-day world that the third wave HCI (Harrison et al., 2007) highlighted, but also opens up for investigations of what happens with trust when “fourth-wave” technologies with their blurred human-machine interfaces start to inhabit it (Frauenberger, 2019). We argue that a viable way to develop deeper understandings about these dimensions of trust is to investigate how people might engage with emerging technologies that do not yet exist in everyday life context, and to implement approaches that are designed to capture the social, situational and cultural aspects of trust from real-world environments.

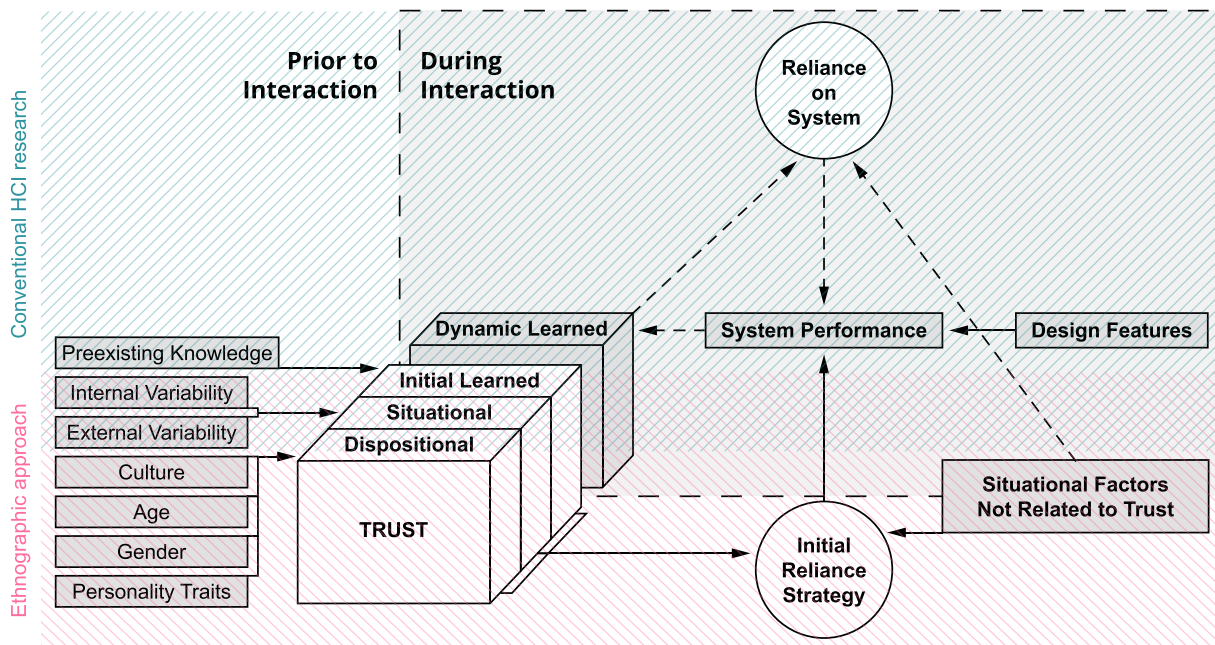


Fig. 8. How ethnography contributes to existing understandings of trust. This diagram builds on Hoff and Bashir's (2015) model presented in Fig. 2 and shows how an ethnographic approach addresses the situational aspects of trust identified as under-theorized by Hoff and Bashir (2015) and Lee and See (2004).

Acknowledgements

This study was financially supported by the Swedish strategic vehicle research and innovation programme (FFI) [grant number 2017-03058] and is a deliverable within the TIC (Trust in Intelligent Cars) project.

References

- Balfe, N., Sharples, S., Wilson, J.R., 2018. Understanding is key: an analysis of factors pertaining to trust in a real-world automation system. *Hum. Factors* 60, 477–495. <https://doi.org/10.1177/0018720818761256>.
- Banks, V.A., Stanton, N.A., 2016. Keep the driver in control: automating automobiles of the future. *Appl. Ergon.* 53, 389–395. <https://doi.org/10.1016/j.apergo.2015.06.020>.
- Banks, V.A., Eriksson, A., O'Donoghue, J., Stanton, N.A., 2018. Is partially automated driving a bad idea? Observations from an on-road study. *Appl. Ergon.* 68, 138–145 (<https://doi.org/10/gcnp2n>).
- Beiker, S., Calo, R., 2012. Legal aspects of autonomous driving. *SSRN J.* <https://doi.org/10.2139/ssrn.2767899>.
- Bødker, S., 2006. When second wave HCI meets third wave challenges. Proceedings of the 4th Nordic Conference on Human-Computer Interaction Changing Roles-NordiCHI'06. Presented at the 4th Nordic Conference. ACM Press, Oslo, Norway, pp. 1–8 <https://doi.org/10.1145/1182475.1182476>.
- Brinkley, J., Posadas, B., Sherman, I., Daily, S.B., Gilbert, J.E., 2019. An open road evaluation of a self-driving vehicle human-machine interface designed for visually impaired users. *Int. J. Hum. Comput. Interact.* 35, 1018–1032. <https://doi.org/10.1080/10447318.2018.1561787>.
- Chng, S., Abraham, C., White, M.P., Skippon, S., 2019. To drive or not to drive? A qualitative comparison of car ownership and transport experiences in London and Singapore. *Transp. Res. Interdiscip. Perspect.* 2, 100030 (<https://doi.org/10/gg68rt>).
- Cohen, T., Stilgoe, J., Stares, S., Akvelken, N., Cavoli, C., Day, J., Dickinson, J., Fors, V., Hopkins, D., Lyons, G., Marres, N., Newman, J., Reardon, L., Sipe, N., Tennant, C., Wadud, Z., Wigley, E., 2020. A constructive role for social science in the development of automated vehicles. *Transp. Res. Interdiscip. Perspect.* 6, 100133 (<https://doi.org/10/gg68rk>).
- Corsín Jiménez, A., 2011. Trust in anthropology. *Anthropol. Theory* 11, 177–196 (<https://doi.org/10/dr459t>).
- Creswell, J.W., 2009. *Research Design: Qualitative, Quantitative, and Mixed Methods Approaches*. 3rd ed. Sage Publications, Thousand Oaks, Calif.
- Diefenbach, S., Hassenzahl, M., 2019. Combining model-based analysis with phenomenological insight: a case study on hedonic product quality. *Qual. Psychol.* 6, 3–26. <https://doi.org/10.1037/qap0000096>.
- Dourish, P., 2001. *Where the Action Is: The Foundations of Embodied Interaction*. MIT Press, Cambridge, Mass.
- Edelmann, A., Stumper, S., Petzoldt, T., 2019. Specific feedback matters—the role of specific feedback in the development of trust in automated driving systems. 2019 IEEE Intelligent Vehicles Symposium (IV). Presented at the 2019 IEEE Intelligent Vehicles Symposium. vol. IV. IEEE, Paris, France, pp. 2223–2228 (<https://doi.org/10/ggkbc>).
- Frauenberger, C., 2019. Entanglement HCI the next wave? *ACM Trans. Comput.-Hum. Interact.* 27, 1–27 (<https://doi.org/10/ggqfcj>).
- Fridman, L., Brown, D.E., Glazer, M., Angell, W., Dodd, S., Jenik, B., Terwilliger, J., Patsek, I., Kindelsberger, J., Ding, L., Seaman, S., Mehler, A., Sipperley, A., Pettinato, A., Seppelt, B., Angell, L., Mehler, B., Reimer, B., 2019. MIT advanced vehicle technology study: large-scale naturalistic driving study of driver behavior and interaction with automation. *IEEE Access* 7, 102021–102038 (<https://doi.org/10/gghppw>).
- Gridling, N., Meschtscherjakov, A., Tscheligi, M., 2012. I need help!: exploring collaboration in the car. Proceedings of the ACM 2012 Conference on Computer Supported Cooperative Work Companion-CSCW'12. Presented at the the ACM 2012 Conference. ACM Press, Seattle, Washington, USA, pp. 87–90 (<https://doi.org/10/ggkbn>).
- Harrison, S., Tatar, D., Sengers, P., 2007. *The Three Paradigms of HCI*.
- Hassenzahl, M., 2010. *Experience Design: Technology for All the Right Reasons, Synthesis Lectures on Human-Centered Informatics*. Morgan & Claypool Publishers, San Rafael, Calif.
- Hengstler, M., Enkel, E., Duelli, S., 2016. Applied artificial intelligence and trust—the case of autonomous vehicles and medical assistance devices. *Technol. Forecast. Soc. Chang.* 105, 105–120. <https://doi.org/10.1016/j.techfore.2015.12.014>.
- Hoff, K.A., Bashir, M., 2015. Trust in automation: integrating empirical evidence on factors that influence trust. *Hum. Factors* 57, 407–434 (<https://doi.org/10/f68kpx>).
- Jian, J.-Y., Bisantz, A.M., Drury, C.G., 2000. Foundations for an empirically determined scale of trust in automated systems. *Int. J. Cogn. Ergon.* 4, 53–71 (<https://doi.org/10/fdt84r>).
- Jung, M.F., Sirkin, D., Gür, T.M., Steinert, M., 2015. Displayed uncertainty improves driving experience and behavior: the case of range anxiety in an electric car. Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems-CHI'15. Presented at the the 33rd Annual ACM Conference. ACM Press, Seoul, Republic of Korea, pp. 2201–2210 <https://doi.org/10/cgvh>.
- Lee, J.D., See, K.A., 2004. Trust in automation: designing for appropriate reliance. *Hum. Factors* 31.
- Lee, J.-G., Kim, K.J., Lee, S., Shin, D.-H., 2015. Can autonomous vehicles be safe and trustworthy? Effects of appearance and autonomy of unmanned driving systems. *Int. J. Hum. Comput. Interact.* 31, 682–691 (<https://doi.org/10/ggkbb3>).
- Lüders, M., Andreassen, T.W., Clatworthy, S., Hillestad, T. (Eds.), 2017. *Innovating for Trust*. Edward Elgar Publishing, Cheltenham, UK.
- Magnusson, T., Anderberg, S., Dahlgren, S., Svensson, N., 2020. Socio-technical scenarios and local practice – assessing the future use of fossil-free alternatives in a regional energy and transport system. *Transp. Res. Interdiscip. Perspect.* 5, 100128 (<https://doi.org/10/gg68rn>).
- Marsh, S., Bibben, M.R., 2005. The role of trust in information science and technology. *Ann. Rev. Info. Sci. Tech.* 37, 465–498 (<https://doi.org/10/dstv33>).
- McLoughlin, S., Prendergast, D., Donnellan, B., 2018. Autonomous vehicles for independent living of older adults—insights and directions for a cross-European qualitative study. Proceedings of the 7th International Conference on Smart Cities and Green ICT Systems. Presented at the 7th International Conference on Smart Cities and Green ICT Systems. SCITEPRESS-Science and Technology Publications, Funchal, Madeira, Portugal, pp. 294–303 (<https://doi.org/10/ggkbb7>).
- Moher, D., Liberati, A., Tetzlaff, J., Altman, D.G., 2009. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. *PLoS Med.* 6 (7).
- Mühl, K., Strauch, C., Grabmaier, C., Reithinger, S., Huckauf, A., Baumann, M., 2019. Get ready for being chauffeured: passenger's preferences and trust while being driven by human and automation. *Hum. Factors* (001872081987289). <https://doi.org/10/ggkcb>.
- Muir, B.M., 1994. Trust in automation: part I. Theoretical issues in the study of trust and human intervention in automated systems. *Ergonomics* 37, 1905–1922 (<https://doi.org/10/c93hnq>).
- Orlovskaya, J., Novakazi, F., Lars-Ola, B., Karlsson, M., Wickman, C., Söderberg, R., 2020. Effects of the driving context on the usage of automated driver assistance systems (ADAS) -naturalistic driving study for ADAS evaluation. *Transp. Res. Interdiscip. Perspect.* 4 (<https://doi.org/10/ggzqrd>).
- Piasek, J., Wierzchowska-Tobis, K., 2018. Acceptance and long-term use of a social robot by elderly users in a domestic environment. 2018 11th international conference on human system interaction (HSI). Presented at the 2018 11th International Conference on Human System Interaction (HSI). IEEE, Gdansk, pp. 478–482 (<https://doi.org/10/ggkj89>).
- Pink, S., Osz, K., Raats, K., Lindgren, T., Fors, V., 2020. Design anthropology for emerging technologies: trust and sharing in autonomous driving futures. *Design Studies* (S0142694X20300247). <https://doi.org/10/ggxp6b>.
- Raats, K., Fors, V., Pink, S., 2019. Understanding trust in automated vehicles. Proceedings of the 31st Australian Conference on Human-Computer-Interaction. Presented at the OZCHI'19: 31st Australian Conference on Human-Computer-Interaction. ACM, Fremantle WA Australia, pp. 352–358 (<https://doi.org/10/ggtprn>).
- Stahl, P., Donmez, B., Jamieson, G.A., 2012. Analysis of the interaction between human operator and automated dispatch in haul truck scheduling. Proceedings of the Human Factors and Ergonomics Society Annual Meeting. 56, pp. 2291–2295 (<https://doi.org/10/ggkksj>).
- Tanner, K., 2018. *Experimental research. Research Methods*. Elsevier, pp. 337–356 <https://doi.org/10.1016/B978-0-08-102220-7.00014-5>.
- Trösterer, S., Wurhofer, D., Rödel, C., Tscheligi, M., 2014. Using a parking assist system over time: insights on acceptance and experiences. Proceedings of the 6th international conference on automotive user interfaces and interactive vehicular applications-AutomotiveUI'14. Presented at the the 6th International Conference. ACM Press, Seattle, WA, USA, pp. 1–8 (<https://doi.org/10/ggkccx>).
- Wintersberger, P., Riener, A., 2016. Trust in Technology as a Safety Aspect in Highly Automated Driving. *i-com*. p. 15 (<https://doi.org/10/ggkcc4>).
- Wintersberger, P., von Sawitzky, T., Frison, A.-K., Riener, A., 2017. Traffic augmentation as a means to increase trust in automated driving systems. Proceedings of the 12th Biannual Conference on Italian SIGCHI Chapter - CHIItaly '17. Presented at the the 12th Biannual Conference. ACM Press, Cagliari, Italy, pp. 1–7 <https://doi.org/10.1145/3125571.3125600>.